

MultiCaM-Vis: Visual Exploration of Multi-Classification Model with High Number of Classes

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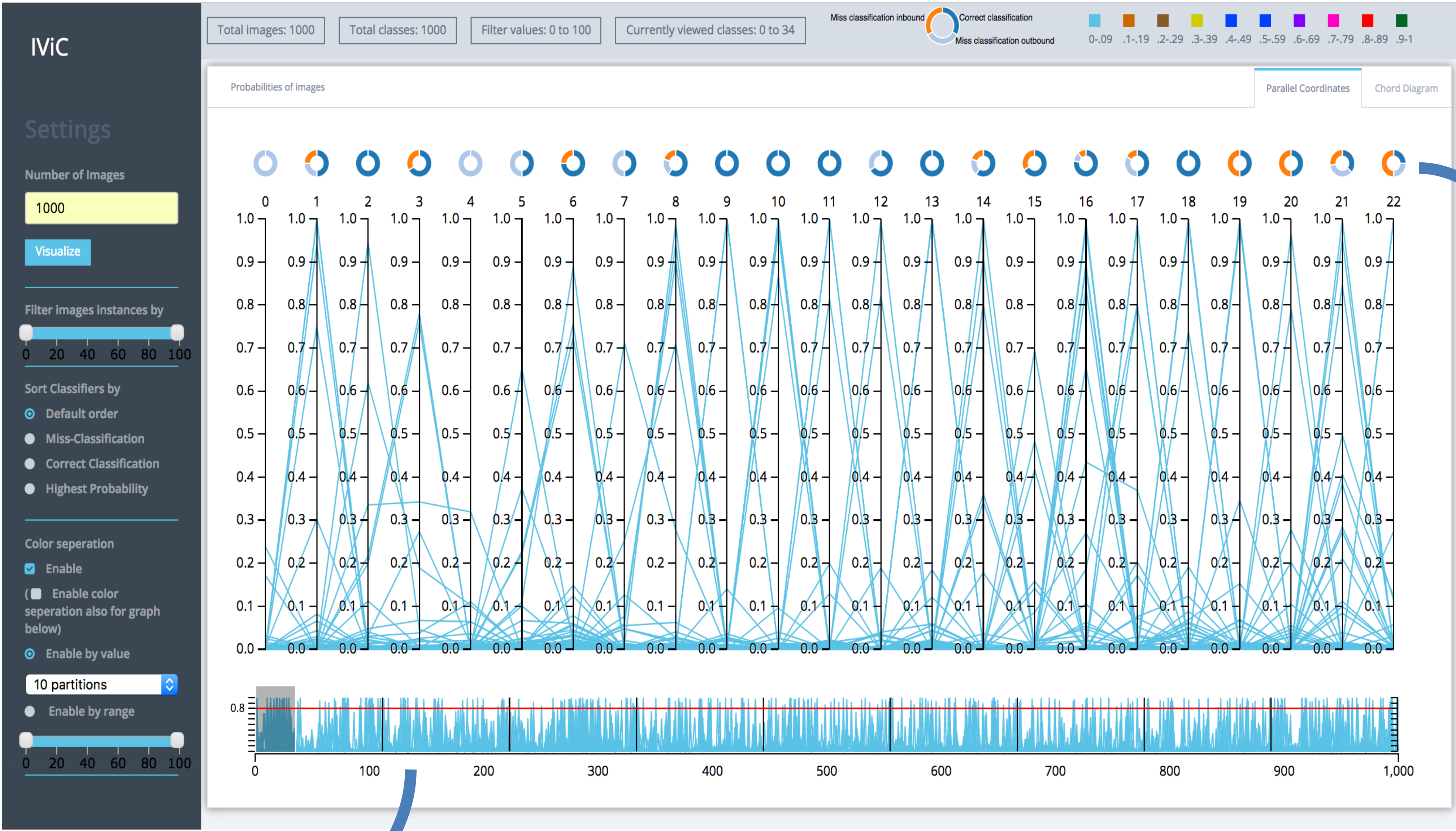


Introduction

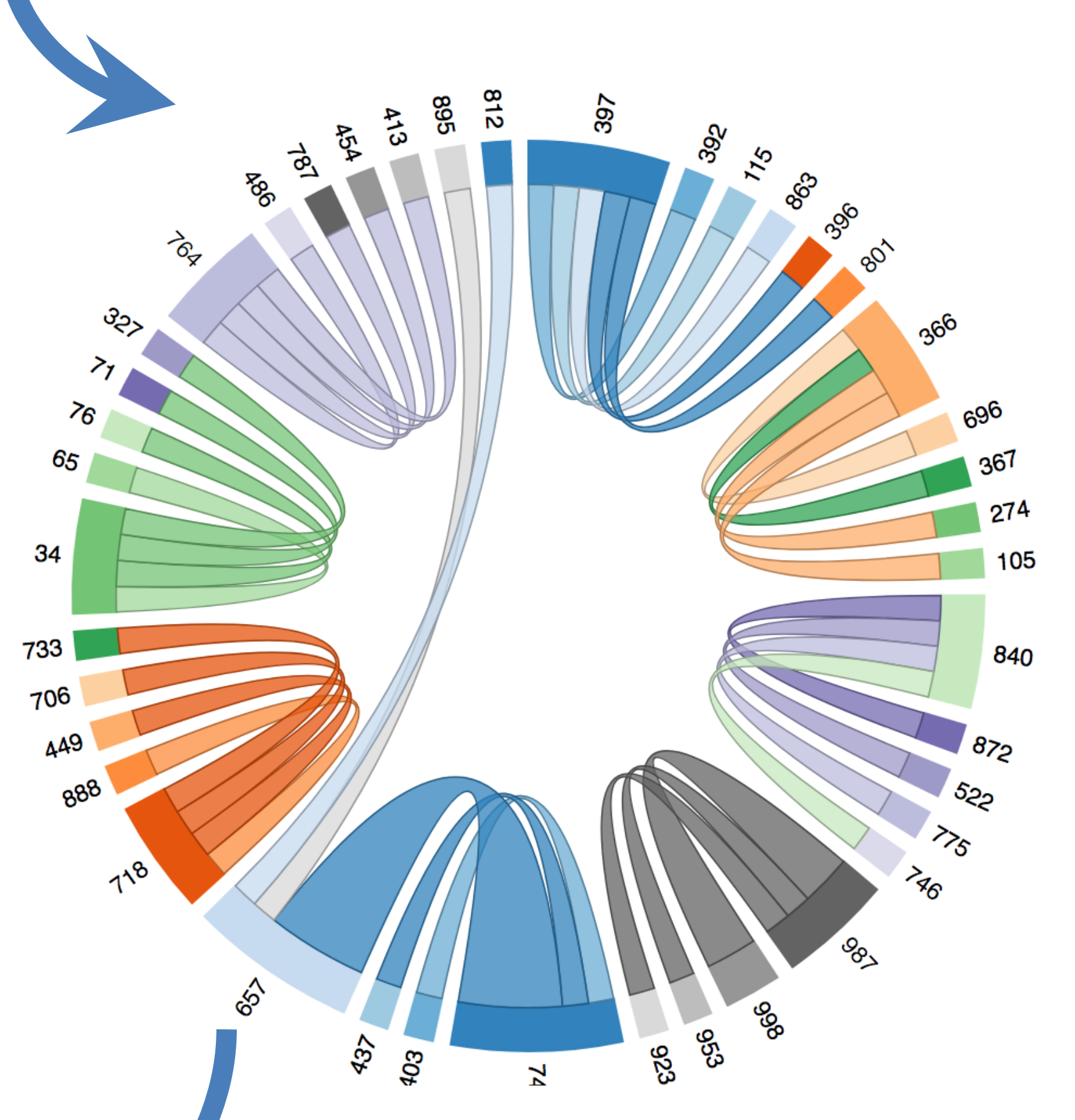
- Problem Statement:** Visual exploration of multi-classification models with large number of classes would help machine learning experts in identifying the root cause of a problem that occurs during learning phase such as miss-classification of instances. Most of the previous visual analytics solutions targeted only a few classes
- Our Solution:** We present our interactive visual analytics tool, called **MultiCaM-Vis**, that provides *overview+detail* style parallel coordinate views and a Chord diagram for exploration and inspection of class-level miss-classification of instances. We also present results of a preliminary user study with 12 participants.
- The Dataset:** **ImageNet dataset consists of a collection of over 15 million** high resolution labelled images, belong to an estimated 22k different categories. We use **ImageNet Large-Scaled Visual Recognition Challenge (ILSVRC) dataset** that is a subset of ImageNet dataset with images belong to 1k different categories. Overall, ILSVRC contains an estimated 1.2 million training images, 50k validation images and 150k testing images. From ILSVRC subset, we use set of validation images with 500 X 500 color photograph in 1K different object classes. The used network in our case consisted of eight learned layers, five convolutional layers and three fully- connected layers. The output of the last fully connected layer (fc8) is a prediction distribution score over the 1000 class labels.

The MultiCaM-Vis Tool

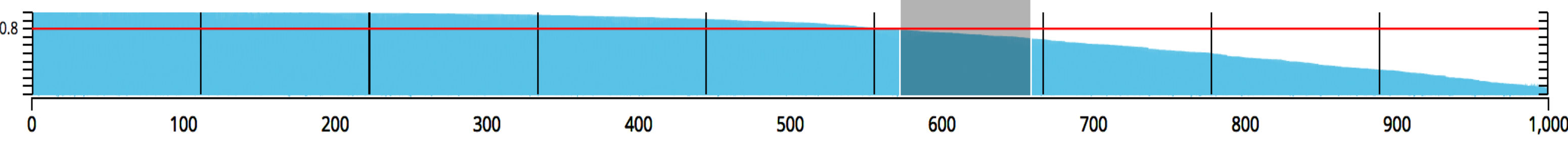
MultiCaM-Vis model exploration view using two parallel coordinates in *Overview+detail* style.



Chord diagram view to inspect incorrect classification cases



Detailed view after sorting the classes based on prediction value.



Mouse hover a particular node (class) highlights the associated chords and nodes as well as images of associated incorrect classification cases.

